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Defining a class with in another

When a class is declared with in another class, the inner class is called as Nested class (ie the inner class) and the outer class is known as Enclosing class. Nested class can be defined in private as well as in the public section of the Enclosing class.

Illustration 14.12 C++ program to illustrate the nested class

```
#include<iostream>
using namespace std;
class enclose
{
private:
        int x;
        class nest
        {
private :
        int y;
public:
int z;
void prn()
{
        y=3;z=2;
        cout<<"\n The product of"<<y<<'*'<<z<<"= "<<y*z<<"\n";
}
}; //inner class definition over
nest n1;
public:
nest n2;
void square()
{
n2.prn(); //inner class member function is called by its object
        x=2;
        n2.z=4;
        cout<<"\n The product of " <<n2.z<<'*'<<n2.z<<"= "<<n2.z*n2.z<<"\n";
        cout<<"\n The product of " <<x<<'*'<<x<<"= "<<x*x;
        }
};              //outer class definition over
int main()
{
enclose e;
        e.square();   //outer class member function is called
}
```